

Consolidation Guide

Scientific Data Analysis v1.9

Contract freeze and app-facing hardening

Stable artifact types, clean errors, P1/P2 consolidation, and expanded app-like regressions

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Translation of the preserved private v1.9 guide at commit `b9f6176`. Original TeX SHA-256: `37247e3ded372906cdca842b400c81b1264a6ecb343226dac5b50a6be8022ae2`. Private source: `skills/scientific-data-maintainer/.cache/active_references/guides/guia_scientific_data_analysis_v1_9.tex`.

Recorded results and closure steps are historical, not current verification. This release guide does not synchronize or modify Scientific Workbench; the app remains outside its scope. Historical commands are retained for traceability: use fresh output directories when reproducing them, without overwriting earlier evidence. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v1.9 is a consolidation release. v1.8 prepared the backend for app consumption; v1.9 reduces debt and freezes the components an app needs to remain stable: artifact type names, app-facing errors, exposure limits, app hints, next actions, and realistic regressions.

v1.9 makes the backend more predictable, its blocks clearer, and its outputs easier to parse before major v2.0 integration. It does not develop or synchronize Scientific Workbench.

Contents

1	Executive overview	1
2	From v1.8 to v1.9	1
3	What remains unchanged	1
4	v1.9 source documents	1
5	P1/P2 inventory	2
5.1	Debt families	2
6	Frozen artifact types	2
6.1	Inherited v1.8 types	3
6.2	Compatible v1.9 additions	3
7	App-facing errors and blocks	3
7.1	Allowed kinds	3
8	App hints and next actions	4
9	Capability exposure	4
10	v1.9 regressions	5
11	Expanded app-like coverage	5
12	Protected v2.0 scope	6
13	Testing a v1.9 capability	6
14	Historical Phase 7 closure	7
15	Toward Phases 8 and 9	7

1. Executive overview

v1.9 is the Contract Freeze and App-Facing Hardening phase. It consolidates the v1.8 work by:

- Classifying inherited P1/P2 debt as v1.9 work, v2.0 deferrals, or not applicable.
- Freezing `artifact_type` names while retaining published v1.8 compatibility.
- Hardening errors, warnings, and `BLOCKED_CONTROLADO` so apps need not interpret tracebacks.
- Reducing risky duplication in envelope, artifact typing, app hint, and next-action helpers.
- Fixing normal, expert, optional, legacy, maintainer, and unexposed modes.
- Expanding app-like regressions with realistic, ambiguous, and broken cases.
- Updating the guide and references to explain consolidation rather than claim v2.0.

The product decision is to prepare the backend for a controlled future integration, without creating Scientific Workbench v2.0 in this release.

2. From v1.8 to v1.9

Area	v1.9 change
Charter	Defines a contract-freeze and app-facing-hardening release.
Debt inventory	Classifies all 59 public capabilities by family, decision, phase, risk, and focused action.
Artifact types	Freezes official names and allowed aliases; prohibits generic types such as <code>json</code> , <code>csv</code> , <code>png</code> , and <code>report</code> .
Errors	Requires <code>errors.kind</code> , <code>errors.message</code> , <code>next_actions</code> , <code>original_modified</code> , and no raw traceback.
App hints	Priority routes emit or document <code>short_summary</code> , <code>severity</code> , <code>preview_artifact_types</code> , <code>tags</code> , and useful actions.
Exposure	Optional, legacy, expert, and maintainer routes stay outside normal mode; the router rejects or demotes them.
Regressions	Adds professional data, documents, inherited notebooks, corrupt containers, time series, sensors, expert astronomy, missing backends, and output conflicts.

3. What remains unchanged

- Scientific Workbench is not edited in v1.9.
- Installed synchronization waits until the final installation phase.
- Published v1.8 artifact types are not renamed without compatibility.
- Domain-specific, legacy, and maintainer tools are not forced into general use.
- UI-driven workflows, job history, preview panels, and a formal app/skill planner remain outside this release.
- Optional backends do not become core dependencies.
- Correct scientific behavior is preserved when improving app-facing output.

4. v1.9 source documents

- `references/v1-9-consolidation-charter.md`: goals, non-goals, phases, and closure.

- `references/v1-9-debt-inventory.md`: actionable P1/P2 inventory for 59 capabilities.
- `references/v1-9-artifact-types-freeze.md`: official artifact types, aliases, and prohibited names.
- `references/v1-9-error-contract.md`: app-facing errors, failures, and blocks.
- `references/v1-9-app-hints-next-actions.md`: priority-route hints and next actions.
- `references/v1-9-exposure-modes.md`: normal, expert, optional, legacy, maintainer, and un-exposed modes.
- `references/v1-9-v2-0-deferrals.md`: protected v2.0 rows.
- `references/v1-9-app-like-extended-regression.md`: expanded app-like coverage.

5. P1/P2 inventory

The inventory covers 59 public capabilities plus two global rows. It classifies app-facing debt as closed, deferred, or explicitly inapplicable; it does not mean that the skill is broken.

Severity	Rows	Interpretation
P1	25	Significant app-facing, optional, artifact-typing, or future UX work.
P2	34	Consolidation, documentation, tests, hints, limits, or regression improvements.
P0	0	No critical release blocker documented in the inventory.

Decision	Rows	Meaning
v1_9	34	Focused consolidation changes.
v2_0	8	Requires Scientific Workbench, UI, or substantial workflow work.
no_procede	17	Close by documenting limits and accurate exposure.

5.1. Debt families

Family	Consolidation scope
artifact_types	Stable typing for outputs, previews, manifests, reports, visual FITS, and sidecars.
app_hints	Short summaries, severity, tags, previews, and useful next actions.
errors	Clean, parseable failures/blocks without raw tracebacks.
exposure_modes	Normal, expert, optional, legacy, and maintainer visibility.
duplicacion	Duplicate helpers affecting envelopes, artifacts, manifests, or next actions.
regressions	Expanded, lightweight, deterministic app-like coverage.

6. Frozen artifact types

Primary reference: `references/v1-9-artifact-types-freeze.md`.

6.1. Inherited v1.8 types

Artifact type	Use
summary_json	Parseable run summary or envelope.
manifest_json	Provenance, input, output, and environment manifest.
report_md	Human-readable Markdown report.
preview_png	Static visual or Quick Look preview.
table_csv	Derived CSV or TSV table.
notebook_ipynb	Generated, inspected, or copy-executed notebook.
log_txt	stdout, stderr, command transcript, or logs.
qa_report	Dedicated QA report.
handoff_bundle	Folder/package for delivery or continuation.
unknown	Explicit fallback when better typing is unavailable.

6.2. Compatible v1.9 additions

Artifact type	Use
fits_product	Derived FITS/FIT/FTS not declared as display-only.
fits_visual	Display-only visual FITS, such as a rendered image RGB_CUBE.
metadata_json	Structured JSON other than the primary summary or manifest.
preview_pdf	PDF used as a preview or preview export.

Prohibited official names are `json`, `csv`, `png`, `pdf`, `markdown`, `fits`, `directory`, `folder`, `preview`, `report`, `manifest`, and `notebook`: they are too generic for the app contract.

7. App-facing errors and blocks

Primary reference: `references/v1-9-error-contract.md`.

A block or failure must provide output that an interface can present directly:

Field	Requirement
status / app_status	Normalizes to <code>BLOCKED_CONTROLADO</code> or <code>FAIL</code> .
errors[].kind	An allowed category, not opaque free text.
errors[].message	A short, actionable human-readable message.
command	Redacted argv/cwd for traceability.
inputs	Inspected inputs, or an explanation that none apply.
typed_artifacts	Present even when empty for early blocks.
next_actions	At least one structured action for a block/failure.
original_modified	<code>false</code> , or explained <code>unknown</code> .
short_summary	Compact summary within <code>app_hints</code> .

7.1. Allowed kinds

Kind	Case
missing_input	Missing path or input.

<code>output_conflict</code>	Output is a directory, has an invalid parent, or is not writable.
<code>corrupt_input</code>	Suffix appears valid, but content is corrupt or truncated.
<code>missing_optional_backend</code>	Optional backend is absent.
<code>invalid_argument</code>	Invalid CLI argument.
<code>unsupported_format</code>	Format/family is unsupported by the selected route.
<code>permission_denied</code>	System permissions prevent reading or writing.
<code>timeout</code>	Backend/subprocess time limit exceeded.
<code>dependency_error</code>	Python dependency or runtime is unavailable.
<code>execution_error</code>	Execution fails after initial validation.
<code>tool_error</code>	Compatible fallback for unclassified legacy messages.

8. App hints and next actions

v1.9 closes Debt B through focused capability profiles, improving explanation without changing the scientific result.

Field	Use
<code>short_summary</code>	Brief sentence for lists, cards, or notifications.
<code>severity</code>	<code>ok</code> , <code>warning</code> , <code>blocked</code> , or <code>error</code> .
<code>preview_artifact_types</code>	Previews the app can highlight.
<code>tags</code>	Route labels such as table, container, astronomy, expert-mode, or QA.
<code>next_actions</code>	Human next step: review columns, correct input, choose a backend, or read a manifest.

`references/v1-9-app-hints-next-actions.md` records 17 closed targets, including:

- General tables: `profile_table.py`.
- General folders: `cross_domain_data_workbench.py`.
- Time series: `timeseries_forecasting_workbench.py`.
- QA/measurement: `physical_qa.py` and `photometry_noise_budget.py`.
- Received notebooks: `coursework_notebook_fidelity_check.py`.
- Expert astronomy: `catalog_workbench.py` `crossmatch-sky`.

9. Capability exposure

v1.9 freezes exposure modes to prevent narrow routes from appearing as ordinary actions.

Mode	Use
<code>normal</code>	Ordinary tables, documents, notebooks, containers, time series, or general QA.
<code>expert</code>	Advanced/domain-specific route requiring technical context.
<code>optional</code>	Optional backend or specific platform; not a core dependency.
<code>legacy</code>	IRAF, <code>fxcor</code> , <code>iSTARMOD</code> , and legacy coursework.
<code>maintainer</code>	Release gates, registry, smoke, synchronization, and probes.
<code>no_exponer</code>	Not a normal action; direct CLI or documented context only.

For general inputs, `scientific_workflow_router.py` recommends normal routes and demotes or rejects sensitive routes with `normal_user_action=false`. The 22 sensitive routes in `references/v1-9-exposure-modes.md` do not become normal v1.9 actions.

10. v1.9 regressions

Gate	Coverage
<code>audit_v1_9_phase0_inventory_regression.py</code>	Charter, 59 rows, P1/P2 counts, decisions, and families.
<code>audit_v1_9_artifact_types_regression.py</code>	Official types, aliases, prohibited names, and Debt A targets.
<code>audit_v1_9_error_contract_regression.py</code>	Clean failures in <code>env_doctor.py</code> , <code>iwork_workbench.py</code> , and <code>duckdb_workbench.py</code> .
<code>audit_v1_9_dedupe_regression.py</code>	Internal helpers and reduced app-facing duplication.
<code>audit_v1_9_app_hints_regression.py</code>	17 targets with hints and next actions.
<code>audit_v1_9_general_p2_regression.py</code>	General/adaptable routes with app-facing P2/P1 work.
<code>audit_v1_9_exposure_modes_regression.py</code>	Router and optional, expert, legacy, and maintainer modes.
<code>audit_v1_9_v2_0_deferrals_regression.py</code>	Eight protected v2.0 rows.
<code>audit_v1_9_more_app_like_regression.py</code>	Additional app-like regression coverage.
<code>audit_v1_9_app_like_extended_regression.py</code>	Realistic, ambiguous, and broken Phase 6 cases.
<code>audit_v1_9_docs_regression.py</code>	References, TeX/PDF guide, extracted text, and rendered pages.

11. Expanded app-like coverage

Phase 6 expands coverage without real private data. Minimum families:

Family	Expected behavior
Professional table	Useful parseable profiling with unchanged originals.
Administrative document	Intake with explicit warnings for weakly supported formats.
Inherited notebook	Preflight blocks or warns about <code>input()</code> and side effects.
Corrupt container	Warning or clean block, not false success.
Time series	Warnings for ambiguous dates, duplicates, or uncertain frequency.
Sensor/measurement	QA for suspicious ranges, NaN/Inf, or questionable units.
Expert astronomy route	Accurate warning/rejection for domain-specific data.
Absent optional backend	<code>BLOCKED_CONTROLADO</code> , not core <code>FAIL</code> .
Output conflict	Clean block with <code>output_conflict</code> .

Recorded main outputs:

```
tmp/v1_9_debt_R_app_like_extended/app_like_extended_regression_summary.json
tmp/v1_9_debt_R_app_like_extended/app_like_extended_regression_report.md
```

12. Protected v2.0 scope

Reference: `references/v1-9-v2-0-deferrals.md`.

Reserved for v2.0:

- Major Scientific Workbench synchronization.
- Complete UI-driven workflows and a formal app planner/skill execution engine.
- Job history, previews, and statuses consumed directly by the app.
- Joint skill/app documentation.
- Interactive column, unit, style, and match selectors.
- Visual confirmations and diffs before editing.
- Complex optional panels for external backends.

Protected capability rows:

- `fits_rgb_batch.py`
- `stilts_workbench.py`
- `radial_velocity_workbench.py` inspect
- `photometric_solution.py`
- `apt_workbench.py`
- `office_roundtrip.py` docx-style-inventory
- `office_roundtrip.py` docx-styled-replace
- `keynote_export.py`

13. Testing a v1.9 capability

Recommended pattern:

1. Create or copy anonymized inputs into a fresh `tmp/` run.
2. Run the router in dry-run mode when input is ambiguous.
3. Invoke the capability with `-summary-json`, `-manifest-json`, or `-run-dir` when available.
4. Validate JSON parsing and accurate `status` / `app_status`.
5. Check `typed_artifacts` against the frozen list.
6. Inspect `app_hints`, `next_actions`, `warnings`, and `errors`.
7. Confirm `original_modified=false`, or explained `unknown`.
8. Classify as `PASS`, `WARNING`, `BLOCKED_CONTROLADO`, `FAIL`, or audit `ROTO`.

Examples:

```
python scripts/audit_v1_9_artifact_types_regression.py
python scripts/audit_v1_9_error_contract_regression.py
python scripts/audit_v1_9_app_like_extended_regression.py
```

14. Historical Phase 7 closure

Phase 7 requires:

- Expected v1.9 references with the required terms.
- This guide in TeX and PDF.
- Successful PDF compilation, extractable text containing key terms, and rendered representative pages or all pages.
- Passing `scripts/audit_v1_9_docs_regression.py`.
- No applicable drift from `scripts/sync_public_surface_docs.py -check`.
- Scientific Workbench unchanged and the installed skill not synchronized.

15. Toward Phases 8 and 9

Phase 8 is the editable-tree release candidate: repeat v1.9 gates, affected v1.8/v1.7/v1.6 regressions, and core smoke. Only Phase 9 may synchronize the installed skill, after a release candidate without FAIL or ROTO.

This guide records the v1.9 documentation state, not final installation:

v1.9 consolidates the backend; v2.0 will synchronize Scientific Workbench.